Quantification: 542_z18
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 13 |
| interior lacuna count | 11 |
| frame-edge lacuna count | 2 |

Interior lacunae only: mean, median and sample SD (n = 11)
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| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 3688.27 | 3684 | 1441.75 | px^2 |
| major_axis_length_px | lacuna shape | 132.275 | 128.431 | 35.9804 | px |
| minor_axis_length_px | lacuna shape | 35.8241 | 35.4482 | 8.8535 | px |
| aspect_ratio | lacuna shape | 3.6839 | 3.6773 | 0.6253 | unitless |
| eccentricity | lacuna shape | 0.9593 | 0.9623 | 0.0137 | unitless |
| solidity | lacuna shape | 0.8643 | 0.893 | 0.0736 | unitless |
| roots_count | headline | 6.1818 | 5 | 5.1927 | count |
| ring_length_r30_px | headline | 399.273 | 403 | 153.861 | px |
| ring_length_r60_px | reported | 1055.18 | 1100 | 357.62 | px |
| owned_length_px | ownership-dependent | 2242.85 | 2419.61 | 1366 | px |
| edge_count | ownership-dependent | 101.364 | 112 | 63.0782 | count |
| mean_edge_length_px | ownership-dependent | 22.5943 | 21.6037 | 4.124 | px |
| perimeter_px | normalised | 349.688 | 337.706 | 124.859 | px |
| ring_area_r30_px2 | normalised | 11901.2 | 12118 | 3231.07 | px^2 |
| ring_area_r60_px2 | normalised | 28359.4 | 29514 | 6840.82 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9834 | 1 | 0.0361 | unitless |
| in_frame_fraction_r60 | normalised | 0.9555 | 1 | 0.0828 | unitless |
| ring_density_r30 | normalised | 0.0327346 | 0.0321236 | 0.00693422 | px^-1 |
| ring_density_r60 | normalised | 0.0365721 | 0.037156 | 0.00790943 | px^-1 |
| roots_per_100px_perimeter | normalised | 1.6497 | 1.6009 | 0.8214 | per 100 px |
| ring_attached_length_r30_px | network v2 | 314.909 | 301 | 163.204 | px |
| ring_attached_length_r60_px | network v2 | 786 | 794 | 380.794 | px |
| ring_length_w_r30_px | network v2 | 444.7 | 443.978 | 175.736 | px |
| ring_length_w_r60_px | network v2 | 1189.75 | 1240.18 | 409.865 | px |
| sholl_crossings_r10 | network v2 | 9.5455 | 9 | 5.2605 | count |
| sholl_crossings_r20 | network v2 | 14.1818 | 14 | 6.29 | count |
| sholl_crossings_r30 | network v2 | 15.8182 | 17 | 5.3818 | count |
| ring_width_mean_r30_px | width | 7.355 | 7.398 | 0.5346 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.00717e+06 | px^2 |
| total_skeleton_length_px | 36672 | px |
| canalicular_length_density_per_px | 0.036411 | px^-1 |
| junction_count | 777 | count |
| junction_density_per_px2 | 0.00077147 | px^-2 |
| lacuna_count | 13 | count |
| lacunae_per_px2 | 1.291e-05 | px^-2 |
| skeleton_component_count | 247 | count |

| | | |
|---|---|---|
| mean_component_length_px | 148.47 | px |
| median_component_length_px | 24 | px |
| field_length_density_w_per_px | 0.0411402 | px^-1 |
| field_density_without_flagged_per_px | 0.0363424 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 372 | px |
| unattached_component_count | 22 | count |
| canalicular_length_density_attached_per_px | 0.036411 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0367804 | px^-1 |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 8.0998 | px |
| width_median_px | 8.2462 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.7703 | px |
| width_pixels_used | 35180 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 602 | 56.2113 | 21.2037 | 2.651 | 0.9261 | 0.6645 | -0.3574 | 16.2608 | 914.042 | True |
| 2 | 5924 | 181.043 | 48.2611 | 3.7513 | 0.9638 | 0.675 | -0.0924 | 129.871 | 638.627 | False |
| 3 | 4438 | 160.681 | 35.4482 | 4.5328 | 0.9754 | 0.8989 | -0.2181 | 173.351 | 382.514 | False |
| 4 | 4342 | 127.796 | 43.8553 | 2.914 | 0.9393 | 0.893 | -0.1983 | 210.3 | 874.108 | False |
| 5 | 5254 | 148.687 | 46.2758 | 3.2131 | 0.9503 | 0.9227 | -0.0688 | 369.252 | 79.7008 | False |
| 6 | 3405 | 124.602 | 35.4524 | 3.5146 | 0.9587 | 0.8958 | -0.0028 | 532.24 | 401.315 | False |
| 7 | 4133 | 149.961 | 37.557 | 3.9929 | 0.9681 | 0.861 | -0.1428 | 589.071 | 924.049 | False |
| 8 | 2572 | 119.294 | 30.2433 | 3.9445 | 0.9673 | 0.8307 | -0.0493 | 570.032 | 785.133 | False |
| 9 | 3684 | 159.337 | 32.9692 | 4.8329 | 0.9784 | 0.8054 | -0.098 | 739.452 | 247.536 | False |
| 10 | 3421 | 128.431 | 34.9252 | 3.6773 | 0.9623 | 0.8774 | -0.2145 | 813.822 | 732.524 | False |
| 11 | 2890 | 110.803 | 33.408 | 3.3167 | 0.9535 | 0.9299 | 0.0034 | 944.816 | 407.904 | False |
| 12 | 508 | 44.3947 | 15.6697 | 2.8331 | 0.9356 | 0.917 | -0.2083 | 990.155 | 937.699 | False |
| 13 | | | | | | | | 1015.14 | 933.902 | True |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 219 | 584 | 872.482 | 48 | 18.1767 | 7.3357 |
| 2 | 20 | 684 | 1477 | 3155.21 | 160 | 19.72 | 7.4739 |
| 3 | 3 | 403 | 1085 | 739.659 | 23 | 32.1591 | 7.5955 |
| 4 | 10 | 488 | 1259 | 3854.17 | 173 | 22.2784 | 7.3773 |
| 5 | 6 | 424 | 1100 | 2973.74 | 152 | 19.5641 | 8.1648 |
| 6 | 5 | 385 | 1117 | 3650.71 | 159 | 22.9604 | 7.1407 |
| 7 | 5 | 535 | 1476 | 1773.79 | 84 | 21.1166 | 7.9782 |
| 8 | 4 | 338 | 972 | 2419.61 | 112 | 21.6037 | 7.3532 |
| 9 | 7 | 365 | 878 | 1474.51 | 69 | 21.3698 | 6.5411 |
| 10 | 5 | 448 | 1323 | 3831.3 | 152 | 25.2059 | 7.5279 |
| 11 | 1 | 223 | 629 | 16.4853 | 1 | 16.4853 | 7.398 |

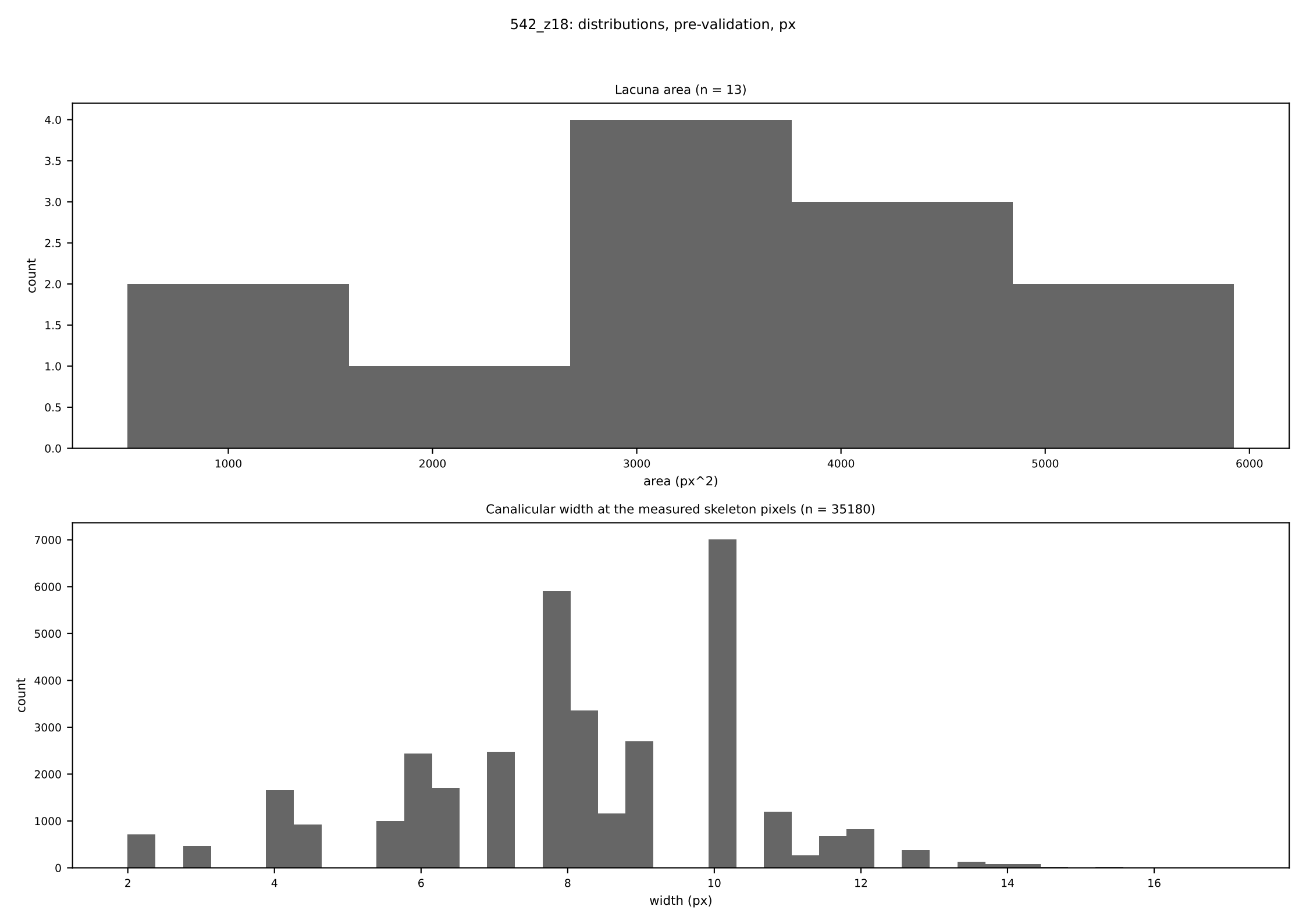
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 12 | 2 | 99 | 291 | 782.21 | 30 | 26.0737 | 6.3545 | |
| 13 | 0 | 0 | 9 | 0 | 0 | 0 | | |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 142.811 | 4575 | 11641 | 0.673 | 0.6207 | 0.0478688 | 0.0501675 | 1.4005 |
| 2 | 625.725 | 16860 | 35430 | 0.9542 | 0.8861 | 0.0405694 | 0.0416878 | 3.1963 |
| 3 | 377.019 | 13152 | 31926 | 1 | 1 | 0.0306417 | 0.0339848 | 0.7957 |
| 4 | 337.706 | 12118 | 29514 | 1 | 1 | 0.0402707 | 0.0426577 | 2.9612 |
| 5 | 371.806 | 13199 | 31872 | 1 | 0.9956 | 0.0321236 | 0.0345131 | 1.6137 |
| 6 | 308.593 | 11379 | 28385 | 1 | 1 | 0.0338343 | 0.0393518 | 1.6203 |
| 7 | 382.877 | 12760 | 30245 | 1 | 1 | 0.0419279 | 0.0488015 | 1.3059 |
| 8 | 306.492 | 10697 | 26160 | 1 | 1 | 0.0315976 | 0.037156 | 1.3051 |
| 9 | 437.262 | 14880 | 35351 | 1 | 1 | 0.0245296 | 0.0248366 | 1.6009 |
| 10 | 326.836 | 11837 | 29271 | 1 | 1 | 0.0378474 | 0.0451983 | 1.5298 |
| 11 | 263.48 | 9968 | 22850 | 0.9798 | 0.8783 | 0.0223716 | 0.0275273 | 0.3795 |
| 12 | 108.77 | 4063 | 10949 | 0.8837 | 0.75 | 0.0243662 | 0.0265778 | 1.8387 |
| 13 | 59.8995 | 971 | 1939 | 0.6158 | 0.5712 | 0 | 0.00464157 | 0 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 212 | 456 | 249.823 | 658.144 | 4 | 7 | 6 |
| 2 | 664 | 1234 | 774.097 | 1686.39 | 22 | 24 | 23 |
| 3 | 238 | 756 | 443.978 | 1207.65 | 8 | 14 | 18 |
| 4 | 365 | 853 | 541.404 | 1440.35 | 10 | 17 | 19 |
| 5 | 301 | 863 | 470.12 | 1240.18 | 5 | 14 | 17 |
| 6 | 294 | 794 | 429.363 | 1277.96 | 8 | 14 | 16 |
| 7 | 471 | 1307 | 601.416 | 1660.87 | 12 | 24 | 20 |
| 8 | 216 | 555 | 367.865 | 1080.25 | 9 | 13 | 15 |
| 9 | 329 | 651 | 415.463 | 998.909 | 12 | 9 | 13 |
| 10 | 390 | 1236 | 498.576 | 1479.74 | 12 | 16 | 20 |
| 11 | 124 | 231 | 245.723 | 700.328 | 5 | 8 | 8 |
| 12 | 72 | 166 | 103.698 | 314.652 | 2 | 3 | 5 |
| 13 | 0 | 0 | 0 | 10.0711 | 0 | 0 | 0 |


542_z18: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 35180)
count
width (px)